

File 32: GT Intern Workstream — Page 1: Rosgen Stream Classification & Field Survey Processing

Hydrologic Engineering & Geomorphic Design Manual Target School: Georgia Institute of Technology (Georgia Tech) School of Civil & Environmental Engineering **Project Area:** Upper Savannah Basin & Chattahoochee River Spawning Headwaters **Supervising Board:** Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)

1. Scientific Principles: Natural Channel Design (NCD)

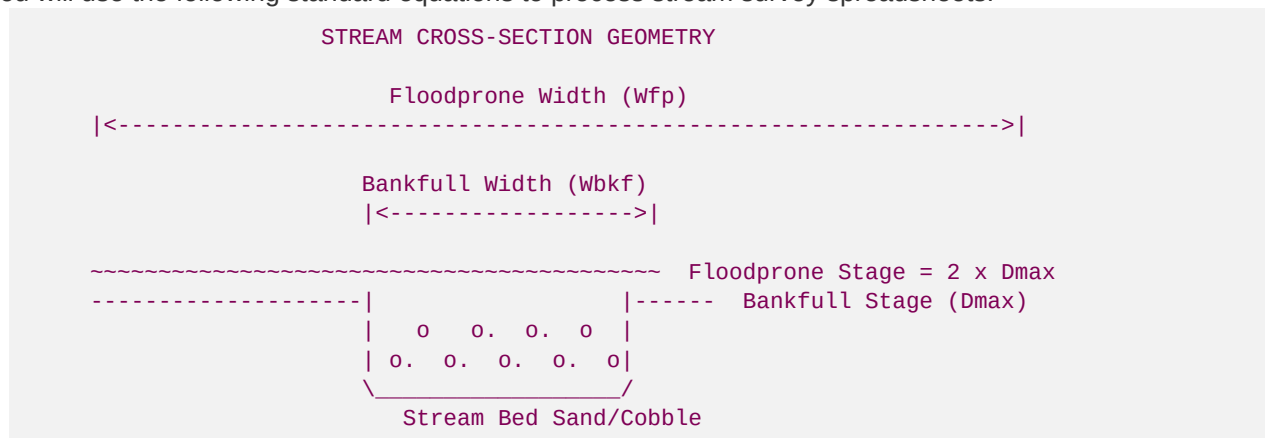
As a Geomorphology & Hydrologic Engineering Intern from Georgia Tech, you are responsible for translating physical stream surveys into stable, self-scouring channel geometries.

We utilize Dave Rosgen's **Natural Channel Design (NCD)** methodology. Degraded mountain streams are often in a state of high geomorphic instability, characterized by rapid lateral erosion (Type F or G channels) due to historic bank modifications. Our design goal is to transition these unstable reaches back to their stable historic geomorphic types (typically **Rosgen B or C channels** in montane ecoregions).

Your first task is to process field-collected cross-section, longitudinal profile, and substrate data to calculate the baseline and design geomorphic parameters.

2. Geomorphic Parameters and Core Equations

You will use the following standard equations to process stream survey spreadsheets:



A. Entrenchment Ratio (ER)

The entrenchment ratio measures the degree of vertical containment of a stream channel within its valley, representing the stream's access to its active floodplain.

$$ER = \frac{W_{fp}}{W_{bkf}}$$

Where:

- W_{bkf} = \text{Bankfull Width (width of channel at bankfull water stage)}.
- W_{fp} = \text{Floodprone Width (width of valley at an elevation equal to twice the maximum bankfull depth, } 2 \cdot D_{max}\text{)}.
- *Classification*: Stable Rosgen B and C channels require $ER \geq 1.4$ (moderately to slightly entrenched). An $ER < 1.4$ indicates severe channel incision (incapable of flooding onto the valley flats, resulting in high shear stress and bank collapse).

B. Width-to-Depth Ratio (WDR)

The width-to-depth ratio indicates the channel's cross-sectional shape and is a primary indicator of sediment transport capacity.

$$WDR = \frac{W_{bkf}}{d_{mean}}$$

Where:

- d_{mean} = \text{Mean bankfull depth } (A_{bkf} / W_{bkf}), where A_{bkf} is the bankfull cross-sectional area.
- *Classification*: Stable high-gradient mountain streams (B-type) typically maintain a WDR between \$12 and \$20. An overwidened stream ($WDR > 25$) experiences a drop in shear stress, causing heavy silt deposition.

C. Sinuosity (K)

Sinuosity measures the degree of stream meandering.

$$K = \frac{L_{channel}}{L_{valley}}$$

Where:

- $L_{channel}$ = \text{Stream channel length measured along the thalweg (deepest point of flow)}.
- L_{valley} = \text{Straight-line valley length between the upstream and downstream points}.

3. Step-by-Step Geomorphic Survey Processing

You will compile field-collected total station or RTK GPS coordinates to classify the stream reach:

Step 1: Process the Cross-Section Data

1. Open the survey CSV sheet containing *Station* (x) and *Elevation* (y) points across the channel cross-section.
2. Identify the **Bankfull Stage** elevation (marked by geomorphic indicators in the field, such as the top of flat depositional benches or changes in vegetation).
3. Calculate the bankfull area (A_{bkf}), bankfull width (W_{bkf}), and maximum depth (D_{max}) relative to this elevation.
4. Calculate $d_{mean} = A_{bkf} / W_{bkf}$.

Step 2: Process the Longitudinal Profile Data

1. Extract the elevation points along the channel thalweg, bankfull stage, and water surface over a length of at least 20 stream widths.
2. Calculate the average channel slope (S):

$$S = \frac{\text{Thalweg Elevation}_{up} - \text{Thalweg Elevation}_{down}}{\text{Stream Length}}$$

3. Identify pool-to-pool spacing and riffle slopes.

Step 3: Classify the Substrate (D_{50})

1. Plot the Wolman Pebble Count data as a cumulative percent finer curve.
2. Extract the D_{50} value (the particle size at which 50% of the substrate is finer).
3. If $D_{50} = 0.5 \text{ mm}$, substrate is sand (Rosgen designation 5). If $D_{50} = 22 \text{ mm}$, substrate is gravel (designation 4). If $D_{50} = 90 \text{ mm}$, substrate is cobble (designation 3).

4. Geomorphic Rosgen Classification Lookup Matrix

Using the parameters calculated above, classify your stream reach according to the following lookup matrix:

Stream Type	Entrenchment Ratio (ER)	Width/Depth (WDR)	Sinuosity (K)	Slope (S)	Geomorphic Description
A	< 1.4 (Highly Entrenched)	< 12	1.0 to 1.2	> 0.04	Steep, entrenched, cascading step-pool stream.
B	1.4 to 2.2 (Moderately Entrenched)	> 12	> 1.2	0.02 to 0.04	Stable, rapids and rapids-dominated steps, low erosion risk.
C	> 2.2 (Slightly Entrenched)	> 12	> 1.2	< 0.02	Meandering pool-riffle channel with broad floodplain.
F	< 1.4 (Entrenched)	> 12 (Very Wide)	> 1.2	< 0.02	Highly unstable, entrenched, severely widening stream bend.
G	< 1.4 (Entrenched)	< 12 (Very Narrow)	> 1.2	0.02 to 0.04	Unstable "gully" channel with active severe bank slumping.

Restoration Goal: If a reach is classified as an eroding **F4** channel, your AutoCAD designs must narrow the bankfull channel and create a new floodplain bench to transition it to a stable **C4** or **B4** channel.